GLOBAL FUEL PRICES AND TAXATION TRENDS

- Project Title: Global Fuel Prices and Taxation Trends: A Comparative Analysis of Petrol and Diesel Prices Across Countries

- Technology Used: Tableau Public, Microsoft Excel/CSV, Data Visualization and Data Analysis

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- **Academic Year: 2026–2027**

INTRODUCTION

PROJECT OVERVIEW

Global Fuel Prices and Taxation Trends is a data visualization and analysis project developed using Tableau Public. The main purpose of this project is to analyze and compare petrol and diesel prices across different countries and understand how fuel taxation and economic factors influence fuel prices.

Fuel is an essential resource for transportation, agriculture, industries, and international trade. Changes in fuel prices can affect transportation costs, the prices of goods, household expenses, and economic development.

Fuel prices are not the same in every country. They depend on several factors, including international crude oil prices, refining costs, transportation expenses, government taxes, subsidies, and exchange rates.

This project uses charts, graphs, filters, and interactive dashboards to present fuel price information in a simple and understandable manner. Tableau helps users explore differences in petrol and diesel prices between countries and identify patterns in the available data.

The project also discusses fuel taxation, including how governments use taxes to generate revenue, influence fuel consumption, and support environmental policies.

PURPOSE

The main purpose of this project is to understand global fuel price variations and examine the role of taxation in determining petrol and diesel prices.

The objectives are:

- To compare petrol and diesel prices across different countries.
- To identify countries with relatively high and low fuel prices.
- To understand how taxation can influence retail fuel prices.
- To analyze differences between petrol and diesel prices.
- To present fuel price information using interactive Tableau dashboards.
- To help users understand international fuel price trends through visual analysis.
- To provide a foundation for further research on fuel affordability, energy policies, and taxation.

IDEATION PHASE

Problem Statement

Fuel prices vary significantly across countries because of differences in crude oil costs, government taxation policies, subsidies, refining costs, and distribution expenses.

Understanding these differences can be difficult when fuel price information is presented in separate tables or complex reports. Users may find it challenging to compare countries, identify price patterns, and understand the possible influence of taxation.

Therefore, the problem is to develop an interactive and user-friendly Tableau dashboard that compares petrol and diesel prices across countries and helps users explore global fuel price variations.

The proposed solution is a data visualization dashboard that organizes fuel price data into meaningful charts and comparisons.

Empathy Map Canvas

An empathy map helps identify the needs and expectations of people who may use the project.

Target users: Students, researchers, analysts, policymakers, and people interested in global fuel prices.

What users think:

- Why are fuel prices different across countries?
- Why is petrol more expensive in some countries?
- How do government taxes affect fuel prices?
- Which countries have relatively affordable fuel?

What users feel:

- Confused by differences in international fuel prices.
- Interested in understanding the reasons behind price changes.
- Concerned about the effect of fuel costs on everyday expenses.

What users see:

- Different petrol and diesel prices in different countries.
- News about changes in crude oil prices.
- Government announcements about taxes and subsidies.
- Charts and reports related to energy markets.

What users say and do:

- Compare fuel prices between countries.
- Search for information about fuel taxes.
- Use charts to understand price differences.
- Discuss how fuel prices affect transportation and living costs.

User needs:

- Simple and understandable visualizations.
- Easy comparison of petrol and diesel prices.
- Interactive country filters.
- Reliable and clearly documented data sources.

Brainstorming

During the brainstorming phase, several ideas were considered for developing the project.

The following features were selected:

- Petrol price comparison by country.
- Diesel price comparison by country.
- Petrol versus diesel price comparison.
- Country-wise fuel price visualization.
- Interactive filters for exploring selected countries and categories.
- Tooltips displaying fuel price details.
- A dashboard combining multiple charts.
- Clear labels, titles, and legends.
- A visual layout that is easy to understand.
- Future integration of historical prices and detailed taxation data.

The final idea was to develop an interactive Tableau dashboard that allows users to explore fuel price differences across countries in one place.

REQUIREMENT ANALYSIS

Customer Journey Map

The customer journey map explains how a user interacts with the dashboard.

Stage 1: Awareness

The user becomes interested in understanding global petrol and diesel prices.

Stage 2: Access

The user opens the Tableau Public dashboard through its published link.

Stage 3: Exploration

The user views the available charts and identifies the countries and fuel categories being compared.

Stage 4: Interaction

The user applies filters, selects countries, and moves the cursor over charts to view additional details.

Stage 5: Comparison

The user compares petrol and diesel prices and observes differences between countries.

Stage 6: Understanding

The user uses the visual information to understand the patterns in the dataset.

Stage 7: Conclusion

The user summarizes the observed differences and considers possible explanations, such as taxes, subsidies, and international market conditions.

Solution Requirement

- ✓ Functional Requirements

The dashboard should:

  - Display petrol and diesel prices for available countries.
  - Allow comparisons between countries.
  - Show fuel prices through suitable charts.
  - Provide interactive filters wherever supported by the dashboard.
  - Display additional information through tooltips.
  - Allow users to explore the dashboard without needing programming knowledge.
  - Be publishable on Tableau Public for online viewing.

- ✓ Non-Functional Requirements

  - Usability: The dashboard should be simple to understand.

- Accuracy: The visualizations should represent the source dataset correctly.
- Performance: Charts and filters should respond smoothly.
- Readability: Titles, labels, legends, and values should be clear.
- Accessibility: The published dashboard should be easy to access through a shared link.
- Reliability: Data sources and units should be documented clearly.

✓ Hardware Requirements

- Computer or laptop.
- Internet connection for publishing and accessing Tableau Public.
- Sufficient memory and storage to run the visualization software.

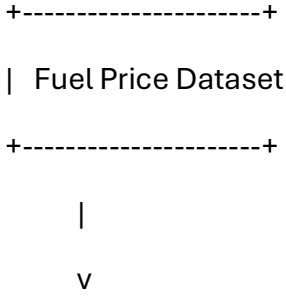
✓ Software Requirements

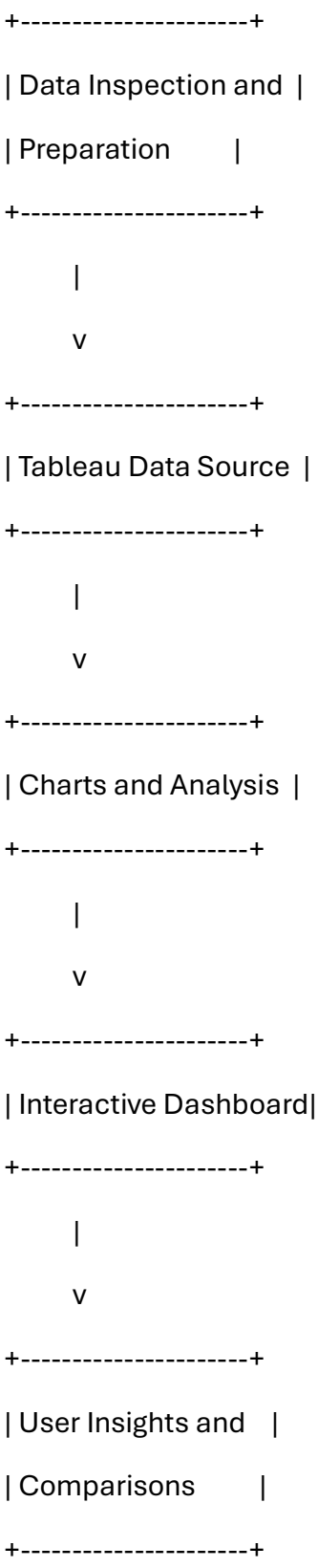
- Tableau Public or a compatible Tableau Desktop version.
- Microsoft Excel or another spreadsheet application.
- CSV or Excel dataset containing fuel prices.
- Web browser.
- Microsoft Word for report preparation.

Data Flow Diagram

The data flow diagram explains how information moves through the project.

Level 0: Overall Data Flow





Explanation:

First, the fuel price dataset is collected. The data is checked for missing values, incorrect formats, duplicate records, and inconsistent units. The cleaned data is then connected to Tableau. Charts are created from the data and combined into a dashboard. Finally, users interact with the dashboard to compare prices and understand the results.

Technology Stack

The following technologies are used in this project.

- Technology| Purpose
- Tableau Public| Creating and publishing interactive dashboards
- Microsoft Excel| Inspecting and preparing tabular data
- CSV/Excel dataset| Storing fuel price information
- Data visualization| Presenting comparisons and patterns
- Web browser| Accessing the published dashboard
- Microsoft Word| Preparing the project report
- GitHub| Storing project documentation and sharing project resources, if used

- Tableau Public: Used to create charts, apply filters, combine worksheets into dashboards, and publish the final visualization online.

- Microsoft Excel: Used to inspect the dataset and check the organization of columns and values.

- CSV or Excel: Used as the input data source for the project.

- Microsoft Word: Used to prepare and format the final project report.

- GitHub: Can be used to store the README file, report, dataset documentation, and other project materials.

PROJECT DESIGN

Problem Solution Fit

The proposed Tableau dashboard addresses the problem of understanding differences in fuel prices across countries.

A simple table may contain many numbers, but it does not always make comparisons easy. Visualizations help users recognize differences more quickly.

The project provides the following solutions:

- Country-wise fuel price comparisons.
- Separate views for petrol and diesel prices.
- A direct comparison of petrol and diesel prices.
- Interactive exploration of available data.
- A consolidated dashboard for presenting the findings.
- Online publication for easy sharing.

The solution is suitable for an educational data visualization project because it demonstrates how raw data can be transformed into useful visual information.

However, price comparison alone cannot prove that taxation caused a particular price difference. To measure the actual impact of taxation, the project would also need country-wise tax rates, tax amounts, or pre-tax and post-tax fuel prices.

Proposed Solution

The proposed solution is to create an interactive dashboard in Tableau Public.

The project follows these steps:

1. Collect the fuel price dataset.
2. Inspect the dataset and identify relevant columns.
3. Connect the dataset to Tableau.
4. Check data types and correct formatting where required.
5. Create the required charts.
6. Add meaningful titles, labels, and tooltips.
7. Add filters for exploring the available data.
8. Combine the charts into a single dashboard.
9. Test the charts and dashboard interactions.
10. Publish the dashboard on Tableau Public.
11. Share the dashboard link and document the project.

The final dashboard helps users compare fuel prices and explore differences between countries.

Solution Architecture

The project architecture consists of five main layers.

1. Data Layer

Contains the fuel price dataset, including country names, petrol prices, diesel prices, units, and dates where available.

2. Data Preparation Layer

Includes data inspection, formatting, validation, and preparation for visualization.

3. Visualization Layer

Contains the Tableau worksheets, charts, calculated fields if required, and visual formatting.

4. Dashboard Layer

Combines the individual charts into one interactive dashboard and provides filters and tooltips.

5. Presentation Layer

The final dashboard is published on Tableau Public and accessed by users through a web browser.

Architecture flow:

Dataset → Data Preparation → Tableau Worksheets → Interactive Dashboard → Tableau Public → User

PROJECT PLANNING & SCHEDULING

Project Planning

The project was divided into the following phases.

Phase| Activity| Expected Output

- Topic selection| Final project topic
- Problem identification| Problem statement
- Dataset collection| Fuel price dataset
- Data inspection| Prepared dataset
- Tableau connection| Connected data source
- Chart creation| Individual visualizations
- Dashboard development| Combined dashboard
- Dashboard testing| Tested charts and filters
- Publishing| Tableau Public dashboard link
- Documentation| Final project report and README

Proposed Schedule

Week 1: Topic selection, problem statement, and data collection.

Week 2: Data inspection and connection to Tableau.

Week 3: Chart creation and dashboard development.

Week 4: Testing, publishing, and report preparation.

The actual schedule may vary depending on dataset preparation, chart requirements, and the time available.

FUNCTIONAL AND PERFORMANCE TESTING

Performance Testing

Testing was carried out to verify whether the charts and dashboard behaved as expected.

The following test cases should be included in the final report based on the tests actually performed.

Test Case| Testing Activity| Expected Result

1| Open the dashboard| Dashboard loads correctly

2| Check chart titles| Titles accurately describe the charts

3| Inspect price labels| Values are displayed in the correct units

4| Hover over a chart| Tooltip displays relevant information

5| Apply a country filter| Relevant chart data updates correctly

6| Apply multiple filters| Charts respond correctly to filter selections

7| Reset filters| The dashboard returns to its default view

8| Check the dashboard layout| Charts and labels remain readable

9| Publish the dashboard| The published version opens correctly

10| Open the shared link| Users can access the dashboard according to its sharing settings

Testing result: Fill in the actual outcome for each test case after testing your published dashboard. A test should be marked as passed only when its expected result is verified.

Performance Evaluation

The dashboard should be evaluated based on:

- Loading time.
- Responsiveness of filters.
- Correctness of displayed values.
- Readability of charts.

- Consistency between the source dataset and the dashboard.
- Accessibility of the published version.

RESULTS

Output Screenshots

The final output of this project is an interactive Tableau dashboard showing comparisons of petrol and diesel prices across countries.

The following screenshots should be inserted in this section.

Figure 1: Petrol Price Comparison by Country

Description: This chart compares petrol prices among the countries included in the dataset. It helps users identify differences in recorded prices.

Figure 2: Diesel Price Comparison by Country

Description: This chart presents diesel prices for the selected countries and allows users to explore price differences.

Figure 3: Petrol versus Diesel Price Comparison by Country

Description: This chart displays petrol and diesel prices together, allowing users to compare the two fuel types within each country.

Figure 4: Interactive Dashboard

Description: The dashboard combines the charts into one interface and allows users to explore the available information through filters and tooltips.

Analysis of Results

The dashboard can be used to investigate the following questions:

- Which countries have the highest and lowest recorded petrol prices in the dataset?
- Which countries have the highest and lowest recorded diesel prices?
- How large is the difference between petrol and diesel prices in each country?
- Do some countries show relatively small differences between petrol and diesel prices?
- How do the results change when different countries or categories are selected?

The answers should be written using the actual values and observations shown in the completed Tableau dashboard.

Interpretation

Differences in fuel prices may be associated with crude oil prices, refining and distribution costs, import dependence, exchange rates, government taxes, and subsidies.

To investigate taxation more directly, the project can compare retail prices with country-wise tax data. The results should not assume that the country with the highest retail fuel price necessarily has the highest tax rate.

ADVANTAGES & DISADVANTAGES

Advantages

1. Provides a clear visual comparison of fuel prices across countries.
2. Makes large datasets easier to explore.

3. Helps users compare petrol and diesel prices.

4. Supports interactive filtering and exploration.

5. Reduces dependence on manually comparing separate tables.

6. Can be shared online through Tableau Public.

7. Provides practical experience in data visualization.

8. Can be extended with historical and taxation data.

9. Helps students understand the use of data in economic analysis.

Disadvantages

1. The quality of the results depends on the accuracy of the dataset.

2. Missing or outdated data can affect comparisons.

3. Different currencies and measurement units can make comparisons misleading if not standardized.

4. Retail fuel prices alone cannot explain the exact effect of taxation.

5. Some countries may have limited or unavailable data.

6. A price comparison does not automatically explain changes over time.

7. Tableau Public is primarily a visualization and publishing platform rather than a complete economic forecasting system.

8. The project may require additional data sources to analyze tax rates, subsidies, and historical trends.

9. Prices should be compared using consistent dates and units wherever possible.

CONCLUSION

The project, Global Fuel Prices and Taxation Trends, demonstrates how Tableau Public can be used to analyze and visualize petrol and diesel prices across different countries.

The dashboard presents fuel price information in a clear and interactive format. Users can compare countries, examine differences between petrol and diesel prices, and explore the information through charts and filters.

The project also highlights the importance of taxation in fuel price analysis. Government taxes and subsidies can influence the final price paid by consumers, while international crude oil markets, refining costs, distribution expenses, and exchange rates also contribute to price differences.

An important conclusion is that retail fuel prices and fuel taxation are related but are not the same measurement. A more complete taxation analysis requires reliable country-wise tax data in addition to retail fuel prices.

Overall, the project demonstrates the value of data visualization in presenting complex information and supporting better understanding of global energy markets.

FUTURE SCOPE

The project can be improved in several ways.

1. Historical Price Analysis: Add multiple years of data to identify changes in fuel prices over time.

2. Taxation Comparison: Include excise duties, VAT, sales taxes, carbon taxes, and other applicable fuel taxes.

3. Pre-Tax and Post-Tax Analysis: Compare fuel prices before and after taxes wherever reliable data is available.

4. Price Trend Charts: Add line charts showing monthly or yearly changes in petrol and diesel prices.

5. Interactive World Map: Use geographical visualizations to display country-wise fuel prices.

6. Economic Comparison: Compare fuel prices with income, inflation, or purchasing power indicators.

7. Subsidy Analysis: Examine how government subsidies influence consumer fuel prices.

8. Affordability Analysis: Estimate the relative cost of fuel for consumers using suitable income or expenditure data.

9. Data Updates: Connect the project to a regularly updated source where practical and permitted.

10. Forecasting: Explore time-series methods to estimate possible future fuel price movements, clearly identifying forecasts as estimates rather than guaranteed prices.

11. Expanded Dashboard: Add more indicators, such as crude oil prices, currency exchange rates, and regional comparisons.

12. Improved Documentation: Maintain a GitHub repository containing the project description, data-source information, screenshots, and instructions for viewing the dashboard.